

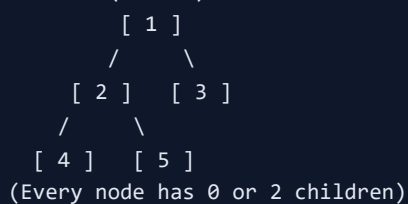
Binary Tree Representation & Types — Data Structures

Definition: A **Binary Tree** is a hierarchical tree data structure in which every node has at most **two children**, referred to as the **left child** and the **right child**.

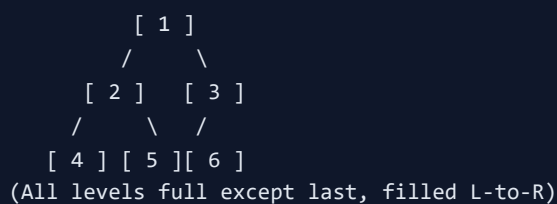
1. Detailed Technical Explanation

1. Types of Binary Trees

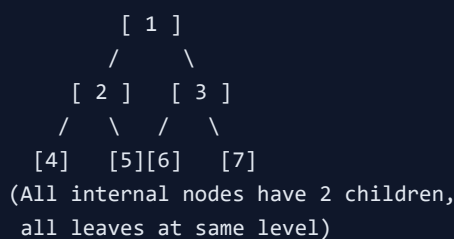
1. FULL (PROPER) BINARY TREE



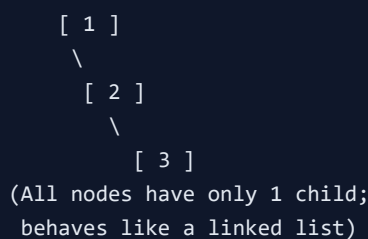
2. COMPLETE BINARY TREE



3. PERFECT BINARY TREE



4. DEGENERATE / SKEWED TREE



- Full (Proper/Strict) Binary Tree:** Every node has either 0 or 2 children. No node has only 1 child.
- Complete Binary Tree:** All levels are completely filled except possibly the last level, which is filled from **left to right** without gaps. (Crucial for Binary Heaps).
- Perfect Binary Tree:** All internal nodes have 2 children and all leaf nodes are at the exact same depth. Total nodes for height h is $2^{(h+1)} - 1$.
- Balanced Binary Tree (AVL/Height-balanced):** The height difference between the left and right subtrees of any node is at most 1 ($|\text{Height}(\text{Left}) - \text{Height}(\text{Right})| \leq 1$).
- Degenerate (Skewed) Tree:** Every internal node has only one child (Left-skewed or Right-skewed). Time complexity degrades to $O(N)$.

2. Binary Tree Representations in Memory

1. Sequential (Array-Based) Representation

For a complete binary tree stored in array **A** (1-indexed):

- Node at index **i** :
- **Left Child:** $2 * i$

- Right Child: $2 * i + 1$

- Parent: $\text{floor}(i / 2)$

2. Linked (Pointer-Based) Representation in C

```
#include <stdio.h>
#include <stdlib.h>

// Structure for a Binary Tree Node
struct TreeNode {
    int data;
    struct TreeNode* left;
    struct TreeNode* right;
};

// Function to create a new tree node
struct TreeNode* createNode(int value) {
    struct TreeNode* newNode = (struct TreeNode*)malloc(sizeof(struct TreeNode));
    newNode->data = value;
    newNode->left = NULL;
    newNode->right = NULL;
    return newNode;
}
```

3. Mathematical Properties of Binary Trees

- Maximum nodes at level i (root at level 0) = 2^i .
- Maximum nodes in binary tree of height h = $2^{(h+1)} - 1$.
- Minimum height of binary tree with N nodes = $\text{ceil}(\log_2(N + 1)) - 1$.
- In any non-empty binary tree, if n_0 is leaf count and n_2 is nodes of degree 2:

$$n_0 = n_2 + 1$$

4. Must-Write Points for Exams

- In a Full Binary Tree, no node has degree 1.
- In a Complete Binary Tree, array storage has zero memory gaps.
- The formula $n_0 = n_2 + 1$ is frequently asked in university proofs and GATE MCQs.

5. Quick Recall Flow

Binary Tree (≤ 2 Children) $\rightarrow$ Types: Full (0/2), Complete (L-to-R), Perfect, Skewed $\rightarrow$ Array ($2i, 2i+1$) vs Linked Node